Structure Factor report

Francisco Vazquez-Tavares

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In this report are the main results of the structure factor analysis.

Characteristic distances in the system and the domains for the fit

The distance range for the pair-wise interaction is defined by the distance at minimum energy, to the cutoff, $r_b \in [\sigma_p, 3/2\sigma_p]$. Then, considering that the center of all particles, central and patch, lies on the same line the distances between the center of the central particles is given by $r_{\text{int}} = 2\sigma_p + r_b$. If we evaluate r_b at the limits, the interaction distance is $r_{\text{int}} \in [1.2, 1.4]$. Finally, if we increase the amount of patchy particles in pair-wise interaction, the distance between central particles is given by

$$d_{\text{int}} = (N-1)\left(2\sigma_p + r_b\right).$$

The result fo this analysis is used to defined the region at small distances. More specifically, the lower limit of this region is set at $N = 2$ and $r_b = 3/2\sigma_p$ and the upper limit is set to $N = 4$ and $r_b = 3/2\sigma_p$. Leading to the following spatial domain,

$$\lambda_1 \in \left[7/2\sigma_p, 21/2\sigma_p\right].$$

Then, we define a second spatial domain, which it's upper limit is set the order of magnitud of the total length of the simulation box, as follows,

$$\lambda_2 \in \left[21/2\sigma_p, L/4\right].$$

About the fit. The experimental data was subjected to an exponential fit

$$f(|\vec{q}|) = \frac{A}{|\vec{q}|^{\alpha}}.$$

The values are the following,

System	Zone	A	α
1	1	0.925179	2.486543
2	1	0.794425	2.08354
1	2	1.264535	2.731100
2	2	0.561453	1.948924

Table 1: Results of the fit.

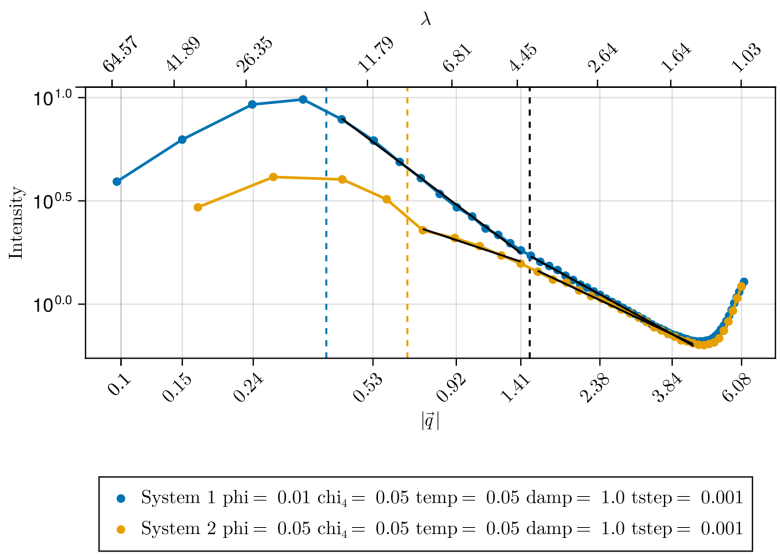


Figure 1: Structure factor of two different systems at the same experiment.